

Multiview Spatial–Temporal Interaction Attention-Based Multivariate Time Series Anomaly Detection for Distributed Industrial Control Networks

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Abstract—Artificial Intelligence-empowered Industrial Control Networks coordinate massive heterogeneous devices and contain multi-node spatial-temporal information. Multivariate Time Series Anomaly Detection (MTS-AD) can discover data-fault behaviors for ensuring the security of distributed networks. However, existing studies tend to rely heavily on single temporal features or neglect the rich spatial-temporal correlations, which leads to the serious underutilization of interactive embeddings between the time and space domains. In this article, a novel Multiview Spatial-Temporal Interaction Attention Network (MSTIA-Net) scheme is proposed for the unsupervised MTS-AD task to better tackle these challenges. MSTIA-Net focuses on jointly modeling the comprehensive spatial-temporal dependencies by means of incorporating complex interactive contents and dynamic relations from multiview patterns. To fully leverage the content-oriented interactions, a spatial-temporal interactions aggregation module is presented to explicitly learn content-aware representations with a parallel-attention mechanism and a low-rank bilinear fusion manner. Simultaneously, considering the potential correlations among different variables as contextual cues, a spatial-temporal correlations learning module is developed

to adaptively capture the relevant context for relation-aware representations. On this basis, both types of aware clues are further integrated by the dual attention-enhanced contrastive reconstruction, which can enrich the cross-aware fusion representations and generate the local and global outputs through a cross-view contrastive learning strategy. Experiments conducted on six benchmark datasets demonstrate the superiority of our MSTIA-Net over state-of-the-art baselines.

Index Terms—Anomaly detection, distributed industrial control networks, multivariate time series, spatial-temporal dependency learning, parallel-attention mechanism.

I. INTRODUCTION

DRIVEN by the continuous progress [1], [2], [3] in intelligent manufacturing systems, the deep integration of massive terminal devices and the widespread deployment of diverse industrial applications have facilitated an explosive growth of multi-source data. Based on a type of distributed architecture, the Industrial Control Networks [4] can coordinate the sensing data and instructions among multiple nodes. To achieve device interconnection and data sharing, the Industrial Control Systems (ICS) [5] have gradually evolved into a core paradigm of distributed networks, which is capable of monitoring industrial data and managing production equipment in the infrastructure scenarios, such as the smart nuclear power plants. Typically, the ICS architecture incorporates enormous amounts of sensory devices from physical, control, supervisory, and corporate layers, as shown in the upper part of Fig. 1. However, some anomalous behaviors in the industrial time-series data may pose the potentially serious threats to the reliability services and new quality security of intelligent systems. Towards the heterogeneous multi-node data generated from ICS operations, Multivariate Time Series Anomaly Detection (MTS-AD) [6], [7], [8] as an unsupervised learning task aims to accurately identify abnormal samples and node malfunctions, which is crucial for maintaining the application security of distributed networks.

Currently, many pioneering studies have made remarkable progress in the MTS-AD task based on deep learning methods, and explored the spatial-temporal information from the following perspectives. 1) Most studies mainly attempted to model the single-domain temporal or spatial dependencies. For example, Su et al. [9] proposed a stochastic Recurrent Neural Network (RNN) to learn the robust representations of

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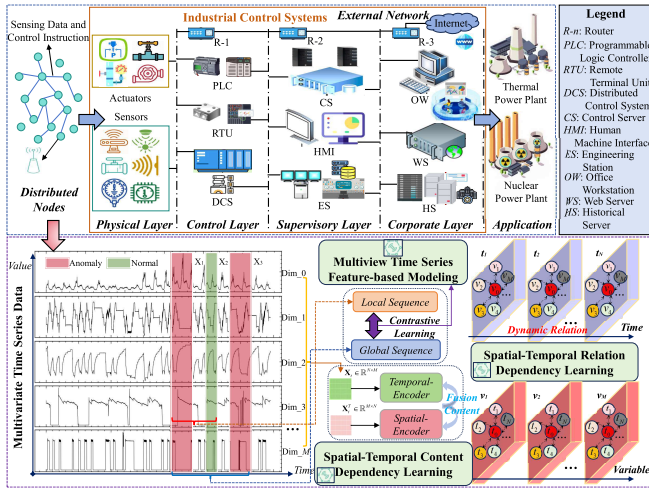


Fig. 1. Illustration of main challenges of the MTS-AD task towards distributed industrial control networks.

temporal dependence. Besides, Deng and Hooi [10] adopted a Graph Neural Network (GNN) to model the spatial relations among multiple variables. These methods heavily rely on the single-domain feature and ignore the interactions of spatial-temporal clues to model the complete feature dependencies. 2) A lot of efforts have also been made to initially explore mining rich spatial-temporal interactions. For instance, Qin et al. [11] presented a multiview contrastive learning scheme to model intra-signal and inter-signal relations, but seriously lacks the exploration of cross-view correlation patterns. Additionally, Ding et al. [12] developed a graph structure learning mechanism to capture the spatial-temporal correlations, and Kang and Kang [13] exploited a variable temporal Transformer to model the spatial-temporal dependencies. However, they were unable to leverage content-aware and relation-aware dependencies in a unified model.

Despite the promising previous works [14], [15], [16], there are still certain limitations that need to be addressed. Existing studies often ignore the deep mining of multi-node relations, and fail to effectively capture the rich spatial-temporal dependencies between different sensing nodes. Considering that massive time-series data are generally obtained from multiple sensing nodes in the ICS platforms, the heterogeneous data distribution tends to show a certain degree of spatial or temporal inconsistency. Due to complex changes in the external distributed environment, it is a tough challenge to fully learn the spatial-temporal correlations of multi-node data. On this basis, we aim to explore an adaptive multi-dependency inference structure to enhance the interactive perception of spatial-temporal clues from each node. The lower part of Fig. 1 illustrates our major motivation, which is to design a dual-channel correlations learning strategy for capturing dynamic dependencies of time-domain and space-domain variables.

In this work, we concentrate on jointly mining the spatial-temporal dependencies of contents and relations from multiview feature patterns, which especially considers the following issues. 1) In general, multivariate local and global information in different positions of sequences can reflect diverse abnormal behavior patterns, and it is necessary to explore uniformly modeling more comprehensive representa-

tions from multiple perspectives. 2) MTS data concurrently depend on the inherent characteristics of both the time-domain and space-domain, so how to fully integrate the spatial-temporal interactive features is critical to better learn the content-based dependency. 3) Since the potential relationships among different variables are relatively dynamic, it can be considered as the contextual clues of the relation-based dependency to adaptively capture spatial-temporal correlations. Accordingly, the core solution for ICS platform is to simultaneously ensure that the two perceptual embeddings are aggregated in a parallel learning strategy.

To address these aforementioned challenges, in this article, we present a MTS-AD approach based on a Multiview Spatial-Temporal Interaction Attention Network (MSTIA-Net). As depicted in Fig. 1, our objective is to model the overall spatial-temporal dependencies of the interactive contents and dynamic relations from the multiview feature patterns in a collaborative manner. In particular, multiview feature extraction module is first utilized to characterize temporal patterns for the local and global features. To jointly explore the rich spatial-temporal information, we then introduce a spatial-temporal interactions aggregation module using a parallel self-attention Transformer and a low-rank fusion method, and simultaneously exploit a spatial-temporal correlations learning module by stacking multilayer Graph Attention Network (GAT). Ultimately, the obtained content-aware and relation-aware representations are both further fused through the local-global cross-attention mechanism, and dual attention-enhanced contrastive reconstruction is developed to generate the dual-stream outputs via a cross-view contrastive learning strategy as an auxiliary means.

In summary, the main contributions can be listed as follows:

- A novel MSTIA-Net scheme is proposed to detect anomalies of time-series data in the distributed industrial networks. MSTIA-Net explicitly models the spatial-temporal dependencies of multiview patterns and learns the interactions between local-global contents and relations.
- To uniformly integrate content-aware representations, a spatial-temporal interactions aggregation module is constructed with the parallel-attention mechanism and low-rank fusion manner. Meanwhile, a spatial-temporal correlations learning module is designed to adaptively capture the dynamic relations among time and variable sequences.
- The dual attention-enhanced contrastive reconstruction is further built to refine cross-aware representation and generate dual-stream outputs through contrastive learning.
- Extensive experiments are conducted to evaluate MSTIA-Net model on six real-world datasets, and results demonstrate that MSTIA-Net achieves significant performance compared to several state-of-the-art methods.

II. RELATED WORK

Recently, benefiting from the powerful ability of deep learning in feature representation and model generalization, MTS-AD task based on the unsupervised deep learning frameworks have received widespread research attention. Existing works on deep learning-based MTS-AD models can be broadly

categorized into two paradigms: prediction-based methods [10], [17], [18] and reconstruction-based methods [9], [19], [20]. The basic idea is how to use the learned normal temporal features to predict future data or reconstruct the original time-series, and the deviation of predicted or reconstructed data from the actual data represents the MTS anomaly degree.

Prediction-based methods predicts future time series data from past data and identifies anomalies based on the prediction errors. For instance, Hundman et al. [17] presented an Long Short-Term Memory (LSTM) based network to forecast future data of input time series in a non-parametric dynamic thresholding manner. Besides, Graph Deviation Network [10] (GDN) designed an attention-based GNN model to learn the relationships and predict the future behaviors of each variable for detecting anomalies in MTS data by temporal deviation.

Reconstruction-based methods reconstructs the original data in a latent space and detects anomalies based on reconstruction errors. For example, Park et al. [21] proposed an LSTM-based Variational Autoencoder (LSTM-VAE) to project the temporal dependencies and reconstruct data distributions. Li et al. [22] combined the Generative Adversarial Network (GAN) and LSTM-RNN models to learn latent interactions. CAE-AD [23] introduced the multi-grained contrastive AE scheme to learn dynamic behaviors of time and frequency domains. Different from the AE-based methods, it can be further divided into the graph-based and attention-based models. For instance, Shi et al. [24] adopted a Graph Convolutional Network (GCN) method to capture the temporal correlations between different variables by an LSTM-based reconstruction. In addition, He et al. [25] presented a variate associated domain adaptation method based on the GDN model to alleviate the cross-domain distribution shift. As for attention-based models, Bai et al. [26] designed a dual attention Transformer scheme to capture the temporal relevancies from global and local patterns. Ma et al. [27] designed a multi-patch Transformer structure, which can model the contextual patterns of both inter-patch and intra-patch view via contrastive learning. Besides, Lin et al. [28] developed a hierarchical context representation learning method to extract the spatial-temporal relations and generate robust multi-scale representations. Qiu et al. [29] presented a unified detection benchmark, which integrates deep learning models on multiple domains.

However, they ignore the multiview interactions of comprehensive spatial-temporal dependencies. In contrast, we are committed to fully learn spatial-temporal representations from interactive contents and dynamic relations in a unified manner.

III. PRELIMINARIES

A. Problem Formulation and Notations

The MTS data generated from Industrial Control Networks can be defined as $\mathcal{T} = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N\}$, which is a sequence of consecutive observations with length N collected at equal-spaced time samples. Generally, the ICS platforms involve multiple sensors, and each time series can contain a variety of signal features from different devices. Given a set of time series metadata $\mathcal{T}_T = \{\mathbf{x}_t\}_{t=1}^T \in \mathbb{R}^{M \times T}$, where T is the sequence length of $\mathcal{T}_T$ and M is the total number of variables.

TABLE I
SUMMARY OF THE MAIN NOTATIONS

Notation	Explanation
$\mathcal{T}_T = \{\mathbf{x}_t\}_{t=1}^T$	A set of multivariate time series with length T .
$\mathbf{X}_t \in \mathbb{R}^{M \times w}$	The t -th sliding slice of inputs with window size w .
$\mathbf{Z}_t^f$	The multiview temporal feature of $\mathbf{X}_t$.
$\mathbf{R}_t^f$	The spatial-temporal content-aware representation.
$\mathbf{g}_i^{l,f}$	The contextual embedding at the l -th GAT layer.
$\mathbf{G}_t^f$	The spatial-temporal relation-aware representation.
$\mathbf{F}_t^f$	The cross-aware fusion representation.
$\hat{\mathbf{X}}_t^{Loc}, \hat{\mathbf{X}}_t^{Glob}$	The local and global reconstructed outputs.

$\mathbf{x}_t = \{x_t^1, x_t^2, \dots, x_t^M\} \in \mathbb{R}^M$ is an M -dimensional vector, which indicates the observation with M variable values at time point t . Considering the dependence of a data point t at a time sample, we use the sliding windows of set value w to obtain a fixed-length input, denoted as $\mathbf{X}_t = \{\mathbf{x}_{t-w+1}, \dots, \mathbf{x}_{t-1}, \mathbf{x}_t\}$. For each $t < w$, we use replication padding manner at the end of window $\mathbf{X}_t$ to ensure that the length of each window is w . The original long-range sequence $\mathcal{T}_T$ is transformed into a sub-sequence set as $\mathcal{X} = \{\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_{N_s}\}$, and each slice is taken as training inputs, where N_s is the number of subsets.

Definition 1 (MTS Anomaly Detection): Given a training set $\mathcal{X}_{train} \in \mathbb{R}^{M \times T}$ and a testing set $\mathcal{X}_{test} \in \mathbb{R}^{M \times \hat{T}}$, the goal of anomaly detection is to predict an anomaly outputs vector $\mathbf{y}_{test} = \{y_1, y_2, \dots, y_{\hat{T}}\} \in \mathbb{R}^{\hat{T}}$, where y_t denotes the anomaly label of each data point t . Here, for an unseen observation $\mathbf{x}_t$, we can use a binary label $y_t \in \{0, 1\}$ to determine whether the t -th time stamp is anomalous or not, where $y_t = 1$ indicates the data point is an anomaly, and $y_t = 0$ otherwise. Due to the scarcity of anomaly labels in ICS data, the MTS-AD is usually regarded an unsupervised task, and we adopt the training set without ground-truth labels for unsupervised learning.

Definition 2 (MTS Anomaly Diagnosis): Given the above training and testing sets, anomaly diagnosis is a qualitative analysis of each dimension based on the detected results. To analyze the severity of anomaly in each anomalous time period, we need to produce the diagnostic outputs $\mathbf{Y}_{test} = \{\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_{\hat{T}}\} \in \mathbb{R}^{M \times \hat{T}}$, where $\mathbf{y}_t \in \{0, 1\}^M$ represents which specific modes of the t -th time stamp are anomalous.

In this work, we aim to tackle unsupervised MTS anomaly detection with an automatic encoder-decoder model $\mathcal{P}$, which is capable of fully leveraging the complex spatial-temporal clues to generate the reconstructed time series as follows:

$$\hat{\mathbf{X}}_t = \mathcal{P}(\mathbf{X}_t | \Theta) = \phi_d \{\phi_e(\mathbf{X}_t; \Theta_e); \Theta_d\}, \quad (1)$$

where $\hat{\mathbf{X}}_t$ denotes the reconstructed outputs, and Θ refers to the set of to-be-learned model parameters. Meanwhile, our unsupervised basic model $\mathcal{P}$ combines an encoding network ϕ_e and a decoding network ϕ_d parameterized by Θ_e and Θ_d . Table I summarizes the descriptions of the main notations.

B. Time Series Data Preprocessing

In real-world industrial scenarios, the collected data is often interfered by some noise due to extreme events. Considering

diverse noise data, we choose a common data preprocessing process with reference to previous works [7], [30] for fair.

1) *Data Normalization*: To enable our model against noise to be robust, we initially normalize the time-series data, both in the training and testing phases. Formally, the specific process of data normalization can be formulated as follows:

$$\tilde{\mathbf{x}}_t \leftarrow \frac{\mathbf{x}_t - \min(\mathcal{T})}{\max(\mathcal{T}) - \min(\mathcal{T}) + \epsilon}, \quad (2)$$

where $\min(\mathcal{T})$ and $\max(\mathcal{T})$ are the minimum and maximum vectors in the original time series $\mathcal{T}$, respectively. In addition, ϵ denotes the small constant vector introduced to prevent the zero-division operation. Notably, we normalize the entire input data to get it within the range of $[0, 1)$.

2) *Handling Noisy Data*: Inspired by randomly adding the Gaussian noise for data augmentation [31], we also use the same noise, which is conformed to multi-type time-series data. To simulate the interference effect of actual noisy data on model generality, we then add the Gaussian white noise with a Signal-to-Noise Ratio (SNR) of 50 dB, which can ensure the presence of noise and overall signal high quality within the typical SNR range of $[45, 55]$ dB. Accordingly, to improve the robustness and generalization of our model, we process the standardized data with the appropriate noise interference and convert it into the sliding time-series windows as follows:

$$\begin{cases} \text{SNR (dB)} = 10 \log_{10} \frac{\sum_{i=1}^N (\tilde{\mathbf{x}}_t)^2}{\sum_{i=1}^N (\epsilon_t)^2}, \\ \mathbf{x}_t = \tilde{\mathbf{x}}_t + \frac{\epsilon_t}{100}, \end{cases} \quad (3)$$

where $\tilde{\mathbf{x}}_t$ and ϵ_t denote the signal and noise vectors, respectively. N is the total number of samples in the MTS data.

IV. PROPOSED METHODOLOGY

A. Overall Framework

To jointly learn the spatial-temporal dependencies of contents and correlations, we aim to develop the spatial-temporal interactive encoding and local-global contrastive decoding structure. The architecture of MSTIA-Net scheme is illustrated in Fig. 2, which involves the following reasoning modules:

- **Multiview Feature Extraction**: It obtains the local and global temporal features of time-series data using two types of independent temporal convolutional structures.
- **Spatial-Temporal Interactions Aggregation**: It initially learns the content-oriented relevant interactions between spatial and temporal dependency embeddings based on a parallel self-attention Transformer structure, and further aggregates the spatial-temporal content-aware representations through a low-rank interaction fusion manner.
- **Spatial-Temporal Correlations Learning**: It leverages multiple spatial-temporal GAT blocks to adaptively capture dynamic correlations among time and variable series for the spatial-temporal relation-aware representations.
- **Dual Attention-Enhanced Contrastive Reconstruction**: It refines the local-branch and global-branch results via cross-attention mechanism, and then adds the contrastive learning strategy as an auxiliary means to generate the cross-aware fusion embeddings for reconstructed outputs.

B. Multiview Feature Extraction

To characterize multi-scale temporal patterns of time-series data from diverse perspectives, we exploit the dual structures of Temporal Convolutional Network (TCN) [32] to extract the local and global temporal features. This module includes two types of TCNs, *i.e.*, causal convolutions as local extractor and dilated convolutions as global extractor. Notably, we construct a skip-layer connection to return the outputs of local branch.

The two main principles of TCN model are to maintain the length of the outputs generated by network equal to the inputs, and to prevent information leakage from the past to the future. To ensure the above two points, the TCN employs the 1D fully convolutional network with regularization operations and pads the input tensor with zeros. Specifically, for each input of MST sub-sequence $\mathbf{X}_t \in \mathbb{R}^{M \times w}$, the basic function of single-layer temporal convolution $\mathcal{F}_k^r(t)$ can be defined as follows:

$$\begin{cases} \mathbf{X}_t^r = \text{Weight-Norm}(\text{Conv1D}_{k,step}^r(\mathbf{W}_c^r, \mathbf{X}_t^{r-1}) + \mathbf{b}_c^r), \\ \mathcal{F}_k^r(t) = \text{Dropout}(\text{LeakyReLU}(\mathbf{X}_t^r)), \quad r \in \{1, \dots, R\}, \end{cases} \quad (4)$$

where $\mathbf{X}_t^r$ is the hidden temporal feature of the r -th layer, and $\text{Conv1D}_{k,step}^r$ denotes 1-dimensional convolution operation with the kernel size of k and the convolution step of $step = 1$. $\mathbf{W}_c^r$ and $\mathbf{b}_c^r$ are the learnable weight matrix and bias term. Besides, $\text{LeakyReLU}(\cdot)$ denotes the LeakyReLU activation function, and R is the number of hidden TCN layers.

1) *Local Feature Encoder*: Benefiting from the potential of short-term time-series forecast, we adopt causal convolutions with each hidden layers of the same length and the zero padding of length $(k - 1)$ to learn the local temporal trends. Formally, the local temporal feature $\mathbf{Z}_t^{\text{Loc}}$ extracted via the causal convolutional network can be expressed as follows:

$$\mathcal{F}_{k_C}^r(t) = (\mathbf{x}_t * f)(t) = \sum_{i=0}^{k_C-1} f(i) \cdot \mathbf{x}(t - i), \quad (5)$$

where $*$ stands for the convolution operation, $f(i)$ represents the i -th filter, and k_C is the size of causal convolution kernel.

2) *Global Feature Encoder*: For high-level temporal feature of long-range sequence as global detection, we apply dilated convolutions to expand the receptive field of each layer and maintain the encoding causality without increasing model complexity. To enhance global feature integration, we design a feedback mechanism via skip-layer connections to re-transmit the local reconstructed outputs predefined as $\hat{\mathbf{x}}_t^{\text{Loc}}$, which can be regarded as implicit clues of inputs $\mathbf{x}_t$. Formally, we can obtain the global temporal feature $\mathbf{Z}_t^{\text{Glob}}$ as follows:

$$\mathcal{F}_{k_D}^r(t) = ((\mathbf{x}_t \oplus \hat{\mathbf{x}}_t^{\text{Loc}}) * f)(t) = \sum_{i=0}^{k_D-1} f(i) \cdot \mathbf{x}(t - d \cdot i), \quad (6)$$

where d is the dilation factor, $\oplus$ denotes the element-wise addition operation, and k_D is the filter size of dilated convolutions. In particular, d determines the previous index, and is equivalent to the fixed step length for skipping between two adjacent filter taps. It corresponds to the j -th power of 2, denoted as $d = 1, 2, 4, \dots, 2^j$. Generally, we set the size of receptive field

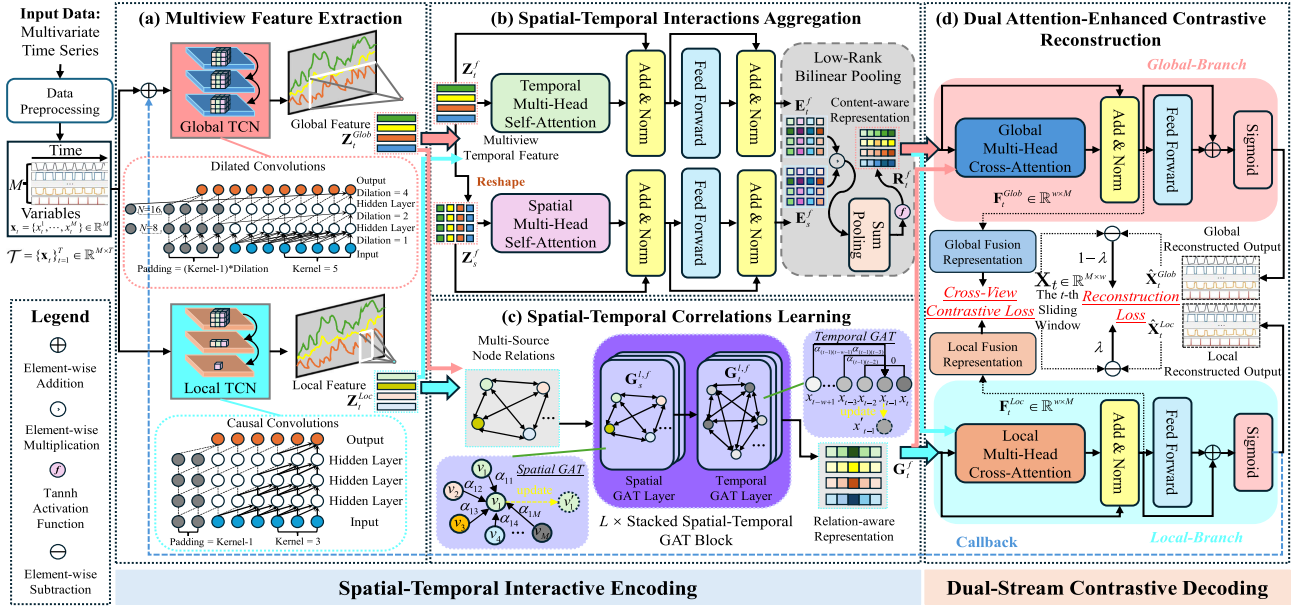


Fig. 2. Overview of MSTIA-Net contains four key modules: (a) Multiview feature extraction, (b) Spatial-temporal interactions aggregation, (c) Spatial-temporal correlations learning, and (d) Dual Attention-enhanced contrastive reconstruction.

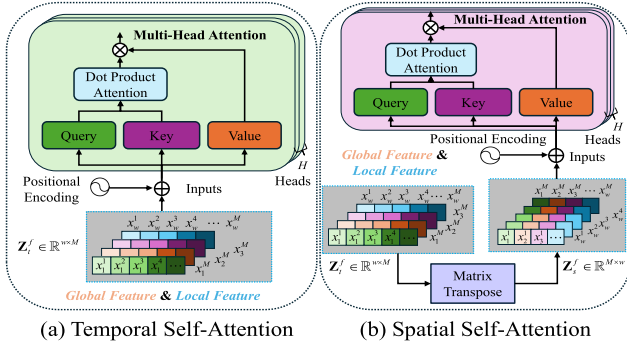


Fig. 3. Illustration of spatial-temporal parallel-attention mechanism, containing (a) Temporal Self-attention transformer and (b) Spatial self-attention transformer architectures.

p to the input length n and the kernel size to k , and solve the number of layers r required for its complete coverage. Under the above conditions, solution process is defined as follows:

$$p = 1 + r \cdot (k - 1), \quad r = \lceil (n - 1) / (k - 1) \rceil. \quad (7)$$

where $\lceil \cdot \rceil$ refers to the rounding up operation. The width of receptive field increases by $d \cdot (k - 1)$ with each additional layer, and is based on the b -exponential expansion, where $d = b^{r-1}$.

For the brevity of this paper, we uniformly use $\mathbf{Z}_t^f \in \mathbb{R}^{w \times M}$ to represent the multiview temporal features of $\mathbf{X}_t$, where $f \in \{Loc, Glob\}$ denotes the local and global temporal features.

C. Spatial-Temporal Interactions Aggregation

Considering the fact that the MTS data depends on various relevant characteristics in the time-domain and space-domain, we present a spatial-temporal interactions aggregation module to simultaneously learn the time-oriented and space-oriented dependencies with a parallel-attention mechanism, and further combine these embeddings in a low-rank fusion manner.

Similar to Transformer [33] model, the structure of spatial-temporal parallel-attention mechanism is illustrated in Fig. 3.

Notably, given the temporal feature of inputs $\mathbf{Z}_t^f \in \mathbb{R}^{w \times M}$, the corresponding spatial feature in the spatial encoder can be first converted into its transposed matrix, denoted as $\mathbf{Z}_s^f \in \mathbb{R}^{M \times w}$. To maintain the temporal position order of entire time-series data and enrich long-term coherent dependencies, we introduce the positional encoding operation, which is incorporated into the original sequence as an additional input. Concretely, the initial temporal embedding is an early combination of input $\mathbf{Z}_t^f$ and positional embedding $\text{PE}(t)$ as follows:

$$\begin{cases} \text{PE}_{(pos, 2i)} = \sin\left(\frac{pos}{10000^{2i/d_h}}\right), \\ \text{PE}_{(pos, 2i+1)} = \cos\left(\frac{pos}{10000^{2i/d_h}}\right), \\ \tilde{\mathbf{Z}}_t^f = \mathbf{Z}_t^f + \text{PE}_{pos}(L_{\{T, S\}}, d_h), \end{cases} \quad (8)$$

where $\tilde{\mathbf{Z}}_t^f$ is the encoded input matrix, $\text{PE}_{pos}(\cdot, d_h) \in \mathbb{R}^{w \times M}$ denotes the embedding matrix of position index in time-series, and $L_{\{T, S\}}$ corresponds to the input length of the temporal or spatial features. Besides, i and d_h are the dimensions of input sub-sequences and the hidden layers in the PE encoding.

1) *Spatial-Temporal Self-Attention Transformer*: The parallel attention mechanism contains two types of forms, and we take the temporal attention as an example. The Temporal Self-Attention Transformer is constructed to model the time-domain interactions using the Multi-Head Attention (MHA) structure, and each self-attention head is obtained as follows:

$$\begin{cases} \text{Self-Attn}_h(\tilde{\mathbf{Z}}_t^f) = \text{softmax}\left(\frac{\mathbf{Q}_T \mathbf{K}_T^T}{\sqrt{d_k}}\right) \mathbf{V}_T, h \in \{1, \dots, H\}, \\ \mathbf{Q}_T = \tilde{\mathbf{Z}}_t^f \mathbf{W}_T^Q, \mathbf{K}_T = \tilde{\mathbf{Z}}_t^f \mathbf{W}_T^K, \mathbf{V}_T = \tilde{\mathbf{Z}}_t^f \mathbf{W}_T^V, \end{cases} \quad (9)$$

where $\text{Self-Attn}_h(\tilde{\mathbf{Z}}_t^f) \in \mathbb{R}^{w \times d_{\text{model}}}$ is the time-oriented dependency embedding of the h -th head, and H is the number of attention heads. $\text{softmax}(\cdot)$ is the Softmax activation function. $\mathbf{W}_T^Q \in \mathbb{R}^{M \times d_q}$, $\mathbf{W}_T^K \in \mathbb{R}^{M \times d_k}$, and $\mathbf{W}_T^V \in \mathbb{R}^{M \times d_v}$ are the learnable projection matrices of the Queries, Keys, and Values,

respectively. In our case, d_q , d_k and d_v denote the dimensions of corresponding projection, where $d_q = d_k = d_v = d_{\text{model}}$.

By integrating multiple attention heads, the spatial-temporal parallel-attention mechanism can be formulated as follows:

$$\begin{aligned} \text{SpatTemp-Attn}(\tilde{\mathbf{Z}}_t^f) &= \text{Multi-Head-Attn}(\tilde{\mathbf{Z}}_t^f) \\ &= \text{Concat}(\text{SA}_1, \dots, \text{SA}_h, \dots, \text{SA}_H) \mathbf{W}_{SA}^O, \end{aligned} \quad (10)$$

where $\mathbf{W}_{SA}^O \in \mathbb{R}^{H \times d_{\text{model}} \times M}$ is the projection matrix, SA_h denotes the h -th head, and $\text{Concat}(\cdot)$ is concatenation operation. The self-attention outputs are fed into the residual connection and position-wise Feed-Forward Network (FFN) as follows:

$$\begin{cases} \mathbf{H}_t^f = \text{Layer-Norm}(\text{Spat\&Temp-Attn}(\tilde{\mathbf{Z}}_t^f) + \tilde{\mathbf{Z}}_t^f), \\ \mathbf{E}_t^f = \text{Layer-Norm}(\text{Feed-Forward}(\mathbf{H}_t^f) + \mathbf{H}_t^f), \end{cases} \quad (11)$$

where $\mathbf{E}_t^f \in \mathbb{R}^{w \times M}$ is the time-oriented dependency embedding, and $\mathbf{H}_t^f$ is the hidden temporal embedding. Furthermore, $\text{Feed-Forward}(\cdot)$ denotes the FFN with two Fully-Connected (FC) layers, expressed as $\text{FFN}(x) = \text{ReLU}(x\mathbf{W}_1 + \mathbf{b}_1)\mathbf{W}_2 + \mathbf{b}_2$, where $\text{ReLU}(\cdot)$ denotes the ReLU activation function. $\text{Layer-Norm}(\cdot)$ is layer normalization operation for outputs.

In the above progress of Temporal Transformer, it targets to learn the temporal dependencies between multiple variables. As for the Spatial Transformer, the input feature $\mathbf{Z}_s^f \in \mathbb{R}^{M \times w}$ is transformed from a transposed form of $\mathbf{Z}_t^f \in \mathbb{R}^{w \times M}$, and we can also derive the space-oriented dependency embedding $\mathbf{E}_s^f \in \mathbb{R}^{M \times w}$ in this similar manner as shown in Fig. 3.

2) *Intermediary Feature-Level Fusion*: Due to the heterogeneous contents of spatial-temporal information, we devote to exploring an intermediary feature-level aggregation strategy via uniformly integrating multiple types of the spatial-temporal cues in a common space. Low-Rank Bilinear Pooling (LRBP) [34] can explicitly model the higher-order interactions fusion between features and reduce computational complexity by means of low-rank constraints. Specifically, time-dependent embedding $\mathbf{E}_t^f$ and space-dependent embedding $\mathbf{E}_s^f$ are unified to obtain the dual-fused results, the content-aware process is implemented through the LRBP fusion manner as follows:

$$\mathbf{R}_t^f = \mathbf{P}^T \tanh \left[(\mathbf{U}^T \mathbf{E}_t^f) \odot (\mathbf{V}^T \mathbf{E}_s^f) \right] + \mathbf{B}, \quad (12)$$

where $\mathbf{R}_t^f \in \mathbb{R}^{w \times M}$ is the spatial-temporal content-aware representation. $\mathbf{U} \in \mathbb{R}^{w \times d_u}$ and $\mathbf{V} \in \mathbb{R}^{M \times d_v}$ are two low-rank matrices used to align the two types of the time-oriented and space-oriented embeddings. $\mathbf{P} \in \mathbb{R}^{d_u \times w}$ and $\mathbf{B} \in \mathbb{R}^{w \times M}$ are the learnable parameters of FC network with a hyperbolic activation function $\tanh(\cdot)$. d_u and d_v are the dimensions of factorized matrices, where $d_u = d_v$ is preset in our condition, and $\odot$ denotes the element-wise multiplication operation.

D. Spatial-Temporal Correlations Learning

Since there are dynamic associations between various time series, we coherently explore spatial-temporal correlations as context cues. Inspired by a GAT [35] structure based on the MHA mechanism, we construct a spatial-temporal correlations learning module with multiple stacked GAT blocks to capture deeper relevant dependencies in both space and time domains.

1) *Spatial Graph Attention Layer*: In terms of the space dimension, the given inputs with M variables at time point t can be formulated as a spatial graph $\mathcal{G}_{x_t}(\mathcal{V}_s, \mathcal{E}_s)$, where the variable set $\mathcal{V}_s \in \mathbb{R}^{M \times w} = \{\mathbf{v}_m | m \in [1, M]\}$ is the node set and the set of spatial relation edges $\mathcal{E}_s \in \mathbb{R}^{M \times M}$ reflects the interactive relation among different variables. In particular, we regard the reformulated matrix of spatial feature $\mathbf{Z}_s^f \in \mathbb{R}^{M \times w}$ as the initial node feature matrix, and each node is denoted as $\mathbf{v}_i \in \mathbb{R}^w$ in our case. Generally, for spatial GAT layer, the attention coefficient of each variable node can be produced via propagating interactive correlations from neighbors as follows:

$$\begin{cases} e_{ij}^f = \text{LeakyReLU}(\mathbf{W}_S^T \cdot \text{Concat}(\mathbf{v}_i^f, \mathbf{v}_j^f)), \\ \alpha_{ij}^f = \frac{\exp(e_{ij}^f)}{\sum_{k \in \mathcal{N}_i} \exp(e_{ik}^f)}, \end{cases} \quad (13)$$

where α_{ij}^f is the correlative degree of the node i to j , $\mathbf{W}_S \in \mathbb{R}^{2w \times 1}$ is the learnable transformation matrix, and $\mathcal{N}_i$ denotes the set of neighbor nodes for node i in a spatial graph.

Subsequently, to better aggregate the context cues of neighbor nodes through multi-level dynamic relations, we can adaptively update the outputs of each graph node by continuously stacking multiple spatial GAT layers as follows:

$$\mathbf{g}_i^{l,f} = \frac{1}{K} \sum_{k=1}^K \left[\text{Sigmoid} \left(\sum_{j \in \mathcal{N}_i} \alpha_{ij}^{k,l,f} \mathbf{W}_S^{k,l,f} \mathbf{v}_j^{l-1,f} \right) \right], \quad (14)$$

where $\mathbf{g}_i^{l,f} \in \mathbb{R}^w$ is the hidden contextual embedding of the l -th GAT layer, $l \in \{1, \dots, L\}$, and L is the total number of stacked GAT blocks. $\mathbf{W}_S^{k,l,f} \in \mathbb{R}^{w \times w}$ is the learnable weight matrix, $\text{Sigmoid}(\cdot)$ denotes the Sigmoid activation function, and K is the number of GAT attention heads. Consequently, we can obtain the space-domain contextual embedding matrix $\mathbf{G}_s^{l,f} = [\mathbf{g}_1^{l,f}, \dots, \mathbf{g}_M^{l,f}]^T \in \mathbb{R}^{M \times w}$, which is regarded as the inputs of the next sequential temporal graph layer in the same spatial-temporal GAT block.

2) *Temporal Graph Attention Layer*: To adaptively capture temporal correlations across different time periods, we define a temporal graph as $\mathcal{G}_{x^s}(\mathcal{V}_t, \mathcal{E}_t)$ with the set of time nodes $\mathcal{V}_t \in \mathbb{R}^{w \times M}$ and the set of temporal relation edges $\mathcal{E}_t \in \mathbb{R}^{w \times w}$, where each node $\{\mathbf{z}_{t-w'} | w' \in [0, w-1]\}$ corresponds to the input time sub-sequence $\mathbf{X}_t$ and $\mathcal{E}_t$ represents the connections between different time nodes at the current moment t and only the other previous moments. Specifically, similar to the above update method of spatial graph, the function of temporal GAT network with multiple stacked GAT layers $\mathcal{F}_{\text{Temp_GAT}}^l(t)$ can be formally performed as follows:

$$\mathbf{G}_t^{l,f} = \mathcal{F}_{\text{Temp_GAT}}^l[\alpha_T^f \mathbf{W}_T^{l,f} (\mathbf{G}_s^{l,f})^T], \quad (15)$$

where $\mathbf{G}_t^{l,f} \in \mathbb{R}^{w \times M}$ is the time-domain contextual embedding matrix of the l -th GAT layer. α_T^f and $\mathbf{W}_T^{l,f} \in \mathbb{R}^{M \times M}$ are the correlative degree and the learnable weight matrix of temporal GAT layer. To simplify the notation of spatial-temporal relation-aware representation with multi-layer GAT adaptive updating, we can use $\mathbf{G}_t^f \in \mathbb{R}^{w \times M}$ to represent the outputs $\mathbf{G}_t^{(L),f}$ of the final GAT block.

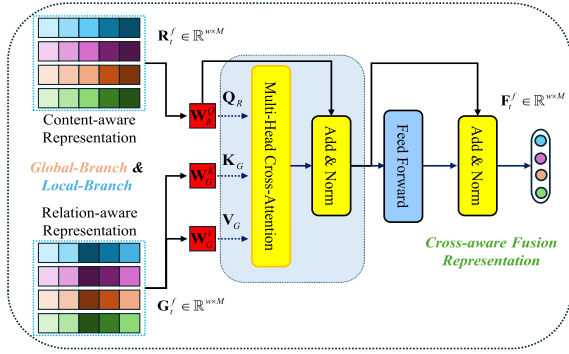


Fig. 4. Illustration of local-global cross-attention mechanism.

E. Dual Attention-Enhanced Contrastive Reconstruction

The content-aware and relation-aware information can sufficiently provide high-level spatial-temporal dependency clues, a dual attention-enhanced contrastive reconstruction module is developed to further ensure the comprehensive fusion representation and dual-stream reconstruction outputs from local and global perspectives, which is refined via the cross-attention mechanism and cross-view contrastive learning strategy.

1) **Local-Global Attention-Based Reconstruction:** To fully aggregate two types of perceptual representations, we devise a local-global cross-attention mechanism, which is illustrated in Fig. 4. The cross-attention block is used to learn cross-aware interactions, and each co-attention head is obtained as follows:

$$\begin{cases} \text{Cross-Attn}_h(\mathbf{R}_t^f; \mathbf{G}_t^f) = \text{softmax} \left(\frac{\mathbf{Q}_R \mathbf{K}_G^T}{\sqrt{d_k}} \right) \mathbf{V}_G, \\ \mathbf{Q}_R = \mathbf{R}_t^f \mathbf{W}_R^Q, \mathbf{K}_G = \mathbf{G}_t^f \mathbf{W}_G^K, \mathbf{V}_G = \mathbf{G}_t^f \mathbf{W}_G^V, \end{cases} \quad (16)$$

where $\text{Cross-Attn}_h(\mathbf{R}_t^f; \mathbf{G}_t^f) \in \mathbb{R}^{w \times d_{\text{model}}}$ is the cross-aware embedding of the h -th head. $\mathbf{W}_R^Q \in \mathbb{R}^{M \times d_q}$, $\mathbf{W}_G^K \in \mathbb{R}^{M \times d_k}$, and $\mathbf{W}_G^V \in \mathbb{R}^{M \times d_v}$ are the learnable projection matrices.

Consequently, we combine multiple co-attention heads, and local-global attention mechanism is described as follows:

$$\begin{aligned} \text{LocGlob-Attn}(\mathbf{R}_t^f; \mathbf{G}_t^f) &= \text{Multi-Head-Attn}(\mathbf{R}_t^f; \mathbf{G}_t^f) \\ &= \text{Concat}(\text{CA}_1, \dots, \text{CA}_h, \dots, \text{CA}_H) \mathbf{W}_{CA}^O, \end{aligned} \quad (17)$$

$$\begin{cases} \mathbf{C}_t^f = \text{Layer-Norm}(\text{Loc\&Glob-Attn}(\mathbf{R}_t^f; \mathbf{G}_t^f) + \mathbf{R}_t^f), \\ \mathbf{F}_t^f = \text{Layer-Norm}(\text{Feed-Forward}(\mathbf{C}_t^f) + \mathbf{C}_t^f), \end{cases} \quad (18)$$

where $\mathbf{F}_t^f \in \mathbb{R}^{w \times M}$ is the cross-aware fusion representation.

Ultimately, for multiview fusion representations, we employ the FFN with two linear layers through a residual connection to generate the reconstructed outputs $\hat{\mathbf{X}}_t^f \in \mathbb{R}^{M \times w}$ as follows:

$$\hat{\mathbf{X}}_t^f = \text{Sigmoid}(\text{Feed-Forward}(\mathbf{F}_t^f) + \mathbf{F}_t^f), \quad (19)$$

where Sigmoid activation is utilized to generate outputs within the range of $[0, 1]$ for matching the normalized input window.

2) **Cross-View Contrastive Learning:** As an auxiliary supervised means, our contrastive learning aims to achieve cross-view alignment between local-global fusion representations, thereby maximizing the consistency of representations across multiple perspectives. Specifically, for the batch size of N_b local-global pairs $\{(\mathbf{F}_i^{Loc}, \mathbf{F}_i^{Glob})\}_{i=1}^{N_b}$, the loss function

for cross-view contrastive learning $\mathcal{L}_{ccl}$ can be defined as follows:

$$\begin{cases} \text{sim}(\mathbf{F}_i^{Loc}, \mathbf{F}_j^{Glob}) = \frac{\text{vec}(\mathbf{F}_i^{Loc}) \cdot \text{vec}(\mathbf{F}_j^{Glob})}{\|\text{vec}(\mathbf{F}_i^{Loc})\| \|\text{vec}(\mathbf{F}_j^{Glob})\|}, \\ \mathcal{L}_{ccl} = -\log \frac{\exp(\text{sim}(\mathbf{F}_i^{Loc}, \mathbf{F}_j^{Glob})/\tau_c)}{\sum_{j=1}^{N_b} \exp(\text{sim}(\mathbf{F}_i^{Loc}, \mathbf{F}_j^{Glob})/\tau_c)}, \end{cases} \quad (20)$$

where $\text{sim}(\cdot, \cdot)$ is the cosine similarity function, $\text{vec}(\cdot)$ denotes the flattening operation, and τ_c is the temperature parameter.

F. Training and Inference

The overall goal is to train our MSTIA-Net model with the reconstruction error and cross-view contrastive terms, and then infer anomaly score based on the predicted and actual values.

1) **Objective Function:** Given the input time series $\mathbf{X}_t$, the model produces dual reconstructed results $\hat{\mathbf{X}}_t^{Loc}$ and $\hat{\mathbf{X}}_t^{Glob}$. We introduce the Mean Square Error (MSE) between the predicted and ground-truth values of two streams in a certain proportion, and reconstruction loss function $\mathcal{L}_{rec}$ is formulated as follows:

$$\begin{aligned} \mathcal{L}_{rec} &= \frac{1}{|\mathcal{X}|} \sum_{\mathbf{x}_t \in \mathcal{X}} (\lambda \|\mathbf{x}_t - \hat{\mathbf{x}}_t^{Loc}\|_F^2 + (1 - \lambda) \|\mathbf{x}_t - \hat{\mathbf{x}}_t^{Glob}\|_F^2), \end{aligned} \quad (21)$$

where $\lambda \in (0, 1)$ is the hyper-parameter to control the weights of local and global flows, and $\|\cdot\|_F$ indicates Frobenius norm.

Accordingly, to jointly optimize the MSTIA-Net, we can present the total objective function $\mathcal{L}_{total}$ by combining the Eq. (20), Eq. (21), and regularization terms as follows:

$$\mathcal{L}_{total} = \mathcal{L}_{rec} + \mu \mathcal{L}_{ccl} + \gamma \|\Theta\|_2^2, \quad (22)$$

where μ and γ are the trade-off parameters to balance the contributions of different losses, and $\|\cdot\|_2$ denotes $\mathcal{L}_2$ -norm.

2) **Meta Learning:** To alleviate the model underfitting on the limited data of real-world scenarios, we utilize the Model-Agnostic Meta Learning (MAML) [36] method to perform data enhancement. In each training phase, for the model $\mathcal{P}(\Theta)$ with parameters Θ in each random batch task $\mathcal{T}_i$, the total gradient update of multiple tasks can be expressed as follows:

$$\begin{cases} \Theta'_i = \Theta - \alpha \nabla_{\Theta} \mathcal{L}_{\mathcal{T}_i}(\mathcal{P}(\Theta)), \\ \Theta = \Theta - \beta \nabla_{\Theta} \sum_{\mathcal{T}_i \sim \mathcal{P}(\mathcal{T})} \mathcal{L}_{\mathcal{T}_i}(\mathcal{P}(\Theta'_i)), \end{cases} \quad (23)$$

where α is the learning rate, and β is the meta-step size. The detailed model training process is summarized in Algorithm 1.

3) **Anomaly Scoring:** In the anomaly inference phase, each sample $\mathbf{x}_t \in \mathcal{X}_{test}$ is input into a well-trained model. Formally, anomaly score of the t -th time stamp can be defined as follows:

$$\text{score}(t) = \|\mathbf{x}_t - \hat{\mathbf{x}}_t^{Glob}\|_2 = \frac{1}{M} \sum_{m=1}^M \sqrt{(x_t^m - \hat{x}_t^m)^2}, \quad (24)$$

where $\text{score}(t)$ is the reconstruction error-based anomaly score at time point t . Since the local representation is regarded as an additional clue for the global representation, we only utilize the global outputs for anomaly scoring. The complete inference process of MSTIA-Net model is shown in Algorithm 2.

Algorithm 1 The Training Procedure of MSTIA-Net

Input: Training data $\mathcal{X}_{train} = \{\mathbf{X}_t\}_{t=1}^{N_s} \in \mathbb{R}^{M \times T}$,
Length of sliding window w , Training epochs I ,
Encoder ϕ_e , Decoder ϕ_d , Hyper-parameters λ, μ, γ .
Output: Trained parameters of model $\Theta = \{\Theta_e, \Theta_d\}$.

- 1: **Initialization:** Randomly initialize all parameters Θ ;
- 2: **for** epoch $i = 1, 2, \dots, I$ **do**
- 3: **for** $t = 1, 2, \dots, N_s$ **do**
- 4: Extract multiview features $\mathbf{Z}_t^f$ with Eq. (4)-(7);
- 5: Calculate the time&space-oriented dependency embeddings $\mathbf{E}_t^f$ and $\mathbf{E}_s^f$ with Eq. (8)-(11);
- 6: Aggregate the spatial-temporal content-aware representations $\mathbf{R}_t^f$ with Eq. (12);
- 7: Update the space&time-domain contextual embeddings $\mathbf{G}_s^{l,f}$ and $\mathbf{G}_t^{l,f}$ with Eq. (13)-(14);
- 8: Learn the spatial-temporal relation-aware representations $\mathbf{G}_t^f$ with Eq. (15);
- 9: Update the cross-aware fusion representations $\mathbf{F}_t^f$ with Eq. (16)-(18);
- 10: Generate the dual-stream reconstructed outputs $\hat{\mathbf{X}}_t^f$ with Eq. (19);
- 11: Calculate the $\mathcal{L}_{ccl}$ and $\mathcal{L}_{rec}$ with Eq.(20)-(21);
- 12: Compute $\mathcal{L}_{total}$ with Eq.(22) and update the Θ ;
- 13: **end for**
- 14: Meta-Learn weights ϕ_e, ϕ_d using a random batch;
- 15: **end for**
- 16: **Return** Optimized model parameters $\Theta = \{\Theta_e, \Theta_d\}$.

Algorithm 2 The Testing Procedure of MSTIA-Net

Input: Testing data $\mathcal{X}_{test} = \{\mathbf{X}_t\}_{t=1}^{\hat{N}_s} \in \mathbb{R}^{M \times \hat{T}}$,
Length of testing sliding window $\hat{w}$,
Encoder ϕ_e , Decoder ϕ_d , Hyper-parameters λ, μ, γ .
Output: Predicted labels of $\mathbf{y}_{test} = \{y_1, y_2, \dots, y_{\hat{T}}\}$.

- 1: Well-trained parameters of the network $\hat{\Theta}_e, \hat{\Theta}_d$;
- 2: **for** $t = 1, 2, \dots, \hat{N}_s$ **do**
- 3: Generate global-stream output $\hat{\mathbf{X}}_t^{Glob}$ with Eq. (19);
- 4: Calculate anomaly score: $\text{score}(t)$ with Eq. (24) ;
- 5: Predict label $y_t = \mathbb{1}(\text{score}(t) \geq \text{POT}(\text{score}(t)))$;
- 6: Produce the list $\mathbf{y}_{test} = \bigvee_t y_t$ with each label y_t ;
- 7: **end for**
- 8: **Return** The predicted anomaly label list $\mathbf{y}_{test} \in \mathbb{R}^{\hat{T}}$.

The inference is to calculate the anomaly score for the inputs processed by the current sliding window, and identify the time point as abnormal label if its score exceeds a certain threshold. We apply the Peak Over Threshold (POT) [37] approach to automatically select anomaly threshold on the validation set.

G. Computational Complexity Analysis

To clearly assess our algorithm performance, we analyze the time complexity of MSTIA-Net based on the data flow. In multiview feature extraction stage, we adopt a dual-branch TCN

TABLE II
STATISTICS OF SIX REAL-WORLD BENCHMARK DATASETS

Datasets	Dimensions	Entities	#Train	#Test	Anomalies (%)
MBA	2	8	100,000	100,000	0.14
MSL	55	3	58,317	73,729	10.72
SMAP	25	55	135,183	427,617	13.13
SWaT	51	1	496,800	449,919	11.98
WADI	123	2	1,048,571	172,801	5.99
SMD	38	4	708,405	708,420	4.16

structure to encode the local and global temporal features. For an input MTS sequence of shape (T, M) , the time complexity of local-global TCNs is $\mathcal{O}(T \cdot R \cdot K_{max} \cdot M)$, where T is the length of time series, M is the number of variables, R denotes the number of TCN layers, and $K_{max} = \max\{k_C, k_D\}$ is the maximum value of convolution kernels. In the interactions aggregation phase, single-view temporal feature is refined by spatial-temporal parallel-attention mechanism, resulting in the complexity of $\mathcal{O}(H \cdot (T^2 \cdot M + T \cdot M^2))$ per Transformer model, and the low-rank bilinear pooling strategy adds the complexity of $\mathcal{O}(D \cdot (T + M) + T \cdot D^2)$, where H is the number of attention heads, and D is the dimension of embedding. In the correlations learning phase, the relation-aware representation is captured by spatial-temporal GAT with the complexity of $\mathcal{O}(L \cdot K \cdot (|\mathcal{V}_s| \cdot T^2 + |\mathcal{E}_s| \cdot T + |\mathcal{V}_t| \cdot M^2 + |\mathcal{E}_t| \cdot M))$, where L denotes the number of stacked GAT blocks, K is the number of GAT attention heads, $|\mathcal{V}_s|$ and $|\mathcal{V}_t|$ are the numbers of variables and time nodes, $|\mathcal{E}_s|$ and $|\mathcal{E}_t|$ are the numbers of spatial and temporal relation edges. In the contrastive reconstruction stage, the cross-aware representation is fused by dual cross-attention mechanism with the complexity of $\mathcal{O}(H \cdot (T^2 \cdot M + T \cdot M^2))$. Therefore, the overall computational complexity of MSTIA-Net model is $\mathcal{O}(T \cdot R \cdot K_{max} \cdot M + H \cdot (T^2 \cdot M + T \cdot M^2) + D \cdot (T + M) + T \cdot D^2 + (L \cdot K \cdot (|\mathcal{V}_s| \cdot T^2 + |\mathcal{E}_s| \cdot T + |\mathcal{V}_t| \cdot M^2 + |\mathcal{E}_t| \cdot M)))$, which simplifies to $\mathcal{O}((L \cdot K \cdot |\mathcal{V}_s| + H \cdot M) \cdot T^2 + (L \cdot K \cdot |\mathcal{V}_t| + H \cdot T) \cdot M^2 + T \cdot D^2)$. This highlights quadratic dependencies on T , M , and D due to attention mechanism and GAT structure.

V. EXPERIMENTS

A. Datasets

To evaluate our proposed MSTIA-Net, we adopt six widely-used datasets for anomaly detection. Table II summarizes the statistics of benchmark datasets. The training set contains only normal data, and abnormal data are labeled in the testing set.

1) *MIT-BIH Supraventricular Arrhythmia Database (MBA)* [38]: It is a popular large-scale dataset for ventricular arrhythmia research, derived from electrical signal recordings in 25 selectively chosen partitions of four patients.

2) *Mars Science Laboratory (MSL) Dataset* [17]: It is a collection of sensor and actuator data with 55 features using the Mars rover itself extracted from the abnormal events of monitoring systems. Notably, we use only the three non-trivial sequences (A4, C2 and T1) identified in the work [8].

3) *Soil Moisture Active Passive (SMAP) Dataset* [17]: It is an expert-labeled database of satellite mission, which comprises the observation data with 25 dimensions.

4) *Secure Water Treatment (SWaT) Dataset* [39]: It is obtained from 51 sensors in the water treatment plant operating for 11 consecutive days, including 7 days of normal operations and 4 days of abnormal conditions under attack scenarios.

5) *Water Distribution (WADI) dataset* [40]: It is a 123-dimensional sensor dataset collected from water distribution networks. Notably, we assign zero values to the missing values.

6) *Server Machine Dataset (SMD)* [9]: It is a 5-week dataset provided by an Internet company, containing resource utilizations from a compute cluster. We utilize only the four non-trivial sequences (machine-1-1, 2-1, 3-2 and 3-7).

In our experiments, we divided the training time-series data into the training set (80%) and validation set (20%). For a fair comparison, we used the same data preprocessing operations as described in preliminaries for all baseline methods.

B. Evaluation Metrics

1) *Anomaly Detection*: To measure the detection performance, we adopted the Precision (Prec), Recall (Rec), F1-Score (F1) and AUC value (the Area Under the ROC Curve) as our main evaluation metrics. For clarity, the values of Prec, Rec, and F1 are calculated as follows:

$$\text{Prec} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \text{Rec} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \text{F1} = \frac{2 \times \text{Prec} \times \text{Rec}}{\text{Prec} + \text{Rec}}, \quad (25)$$

where TP, FP, and FN are the number of true positive, false positive, and false negative results, respectively. Besides, we also employed a common point-adjust [41] method to calculate evaluation metrics. To further verify model performance under limited scale dataset, we also measure the AUC and F1 scores with the same data splitting ratio by training all models with only 20% of the training data, termed as ACU* and F1*.

2) *Anomaly Diagnosis*: To estimate the diagnosis performance, we adopted two main evaluation metrics: HitRate@P% and NDCG@P% [9]. The former is to determine how many ground-truth dimensions are hit in the top-ranked candidates via ranking the anomaly scores predicted by model, and the latter is to measure the Normalized Discounted Cumulative Gain (NDCG) considered with the same number of top predicted candidates. Notably, P% denotes the percentage of ground-truth dimensions for each timestamp.

C. Implementation Details

For the local feature encoder, we applied a causal convolutional network, where the number of hidden TCN layers is set as $R = 3$ and the kernel size is $k_C = 3$. As for the global feature encoder, we employed a dilated convolutional network with three hidden layers, dilation factor $d \in \{1, 2, 4\}$ and convolution kernel $k_D = 5$. We used the grid-search strategy to determine the optimal values for all hyper-parameters. In the spatial-temporal self-attention and local-global cross-attention mechanism, we both adopted the MHA mechanism, where the number of attention heads is set as $H = 3$, and the dimension of projection space is set as $d_{\text{model}} = 64$. In the correlations learning stage, we constructed the multi-level stacked spatial-temporal GAT blocks, where the numbers of layers and attention heads are set as $L = 4$ and $K = 4$. For all

datasets, we set the size of sliding window as $w = 10$, and the dropout operation with a fixed value of 0.2. All experiments are implemented by PyTorch-1.11.0 framework and performed on the environment with a single NVIDIA RTX 3090 GPU.

For the optimization of our proposed MSTIA-Net model, we employed the AdamW method as training optimizer with an initial learning rate of $1e^{-2}$ for SMAP and SWaT, $1e^{-3}$ for MBA and MSL, $1e^{-4}$ for WADI and SMD, the meta learning rate of $\alpha = 2e^{-2}$, and the weight decay of $\gamma = 1e^{-5}$. The step-scheduler strategy is used to adjust the optimizer with step size of 5 and learning decay rate of 0.9. As for the loss function, the trade-off parameters of reconstruction loss in Eq. (21) and total objective function in Eq. (22) are experimentally set as $\lambda = 0.7$ and $\mu = 1e^{-3}$, and the fixed parameter of cross-view contrastive learning in Eq. (20) is empirically set as $\tau_c = 0.5$.

In training process, we use an early-stopping criteria to prevent overfitting and the batch size is 256. For POT parameters, the coefficient is set as $1e^{-4}$ for all datasets, and low quantile is set as $7e^{-2}$ for SMAP, $1e^{-2}$ for MSL, and $1e^{-3}$ for others.

D. Comparison With State-of-the-Art Methods

To validate the effectiveness of our proposed MSTIA-Net scheme on the both anomaly detection and diagnosis tasks, we mainly compared our model with the following state-of-the-art baselines, including the prediction-based, reconstruction-based, graph-based, and attention-based methods.

- **LSTM-NDT** [17]: An LSTM-based network that interprets predictions with a dynamic error threshold strategy.
- **DAGMM** [42]: An autoencoding method that combines the Deep AE and Gaussian Mixture Model to estimate the density of representations in the low-dimensional space.
- **OmniAnomaly** [9]: A RNN-based model that uses the VAE and GRU to generate reconstruction probabilities.
- **MSCRED** [19]: A ConvLSTM-based network that captures the temporal patterns and inter-sensor correlations from multi-scale signature matrices in different statuses.
- **MAD-GAN** [22]: An LSTM-based method that integrates the GAN model to learn the temporal distributions using the prediction errors and discriminator loss.
- **USAD** [43]: An autoencoding-based approach that employs two independent AEs and GAN model to enhance performance and stable with adversarial training strategy.
- **MTAD-GAT** [7]: A graph-based framework that utilizes two parallel GATs to capture correlations between different timestamps and signals through multi-task learning.
- **CAE-M** [31]: An autoencoding method that builds a convolutional AE and memory network with the attention-based BiLSTM method and an autoregressive model.
- **GDN** [10]: A graph-based model that uses a GNN model to learn pairwise correlations between multiple sensors by graph attention mechanism and then detects anomalies according to the predicted deviations from node relations.
- **TranAD** [44]: An attention-based network that incorporates Transformer with the adversarial training and self-conditioning process to improve the model stability.

TABLE III

PERFORMANCE COMPARISON OF OUR MSTIA-NET WITH STATE-OF-THE-ART METHODS ON THE SIX COMPLETE DATASET. THE BEST RESULTS OF AUC AND F1 SCORE ARE HIGHLIGHTED IN BOLD, WHILE SUBOPTIMAL RESULTS ARE UNDERLINED

Method	Sources	MBA				MSL				SMAP			
		Prec	Rec	AUC	F1	Prec	Rec	AUC	F1	Prec	Rec	AUC	F1
LSTM-NDT	KDD'18	0.9206	0.9717	0.9779	0.9455	0.6287	1.0000	0.9531	0.7720	0.8522	0.7325	0.8601	0.7877
DAGMM	ICLR'18	0.9474	0.9900	0.9857	0.9682	0.7362	1.0000	0.9715	0.8481	0.8068	0.9890	0.9884	0.8887
OmniAnomaly	KDD'19	0.8560	1.0000	0.9569	0.9224	0.7847	0.9923	0.9781	0.8764	0.8129	0.9418	0.9888	0.8727
MSCRED	AAAI'19	0.9271	1.0000	0.9798	0.9622	0.8911	0.9861	0.9806	0.9362	0.8174	0.9215	0.9820	0.8663
MAD-GAN	ICANN'19	0.9395	1.0000	0.9835	0.9688	0.8515	0.9929	0.9861	0.9168	0.8156	0.9215	0.9890	0.8653
USAD	KDD'20	0.8954	0.9990	0.9702	0.9444	0.7949	0.9912	0.9795	0.8822	0.7481	0.9628	0.9890	0.8419
MTAD-GAT	ICDM'20	0.9019	1.0000	0.9720	0.9483	0.7918	0.9825	0.9890	0.8769	0.7992	0.9992	0.9846	0.8882
CAE-M	TKDE'21	0.8443	0.9998	0.9662	0.9155	0.7752	1.0000	0.9904	0.8734	0.8194	0.9568	0.9902	0.8828
GDN	AAAI'21	0.8833	0.9893	0.9529	0.9333	0.9309	0.9893	0.9815	0.9592	0.7481	0.9892	0.9865	0.8519
TranAD	VLDB'22	0.9576	1.0000	0.9886	0.9783	0.9037	0.9999	0.9915	0.9493	0.8104	0.9998	0.9887	0.8953
MUTANT	KBS'23	0.9632	0.9865	0.9828	0.9747	0.9268	0.9835	0.9847	0.9543	0.7352	0.9826	0.9768	0.8415
DCdetector	KDD'23	0.9585	0.9999	0.9872	0.9783	0.9172	0.9969	0.9879	0.9554	0.8216	0.9862	0.9909	0.8964
DTAAD	KBS'24	0.9608	1.0000	<u>0.9896</u>	0.9800	0.9038	0.9999	0.9918	<u>0.9495</u>	0.8220	0.9999	0.9911	0.9023
MSTIA-Net	Ours	0.9638	0.9999	0.9903	0.9815	0.8986	0.9999	0.9924	0.9466	0.8267	0.9999	0.9897	0.9051

Method	Sources	SWaT				WADI				SMD			
		Prec	Rec	AUC	F1	Prec	Rec	AUC	F1	Prec	Rec	AUC	F1
LSTM-NDT	KDD'18	0.777	0.5108	0.7139	0.6166	0.0137	0.7822	0.6720	0.0270	0.9735	0.8439	0.9670	0.9041
DAGMM	ICLR'18	0.9932	0.6878	0.8435	0.8127	0.0759	0.9980	0.8562	0.1411	0.9102	0.9913	0.9953	0.9490
OmniAnomaly	KDD'19	0.9781	0.6956	0.8466	0.8130	0.3157	0.6540	0.8197	0.4259	0.8880	0.9984	0.9945	0.9400
MSCRED	AAAI'19	0.9991	0.6769	0.8432	0.8071	0.2512	0.7318	0.8411	0.3740	0.7275	0.9973	0.9920	0.8413
MAD-GAN	ICANN'19	0.9592	0.6956	0.8462	0.8064	0.2232	0.9123	0.8025	0.3587	0.9990	0.8439	0.9932	0.9149
USAD	KDD'20	0.9977	0.6879	0.8460	0.8143	0.1874	0.8297	0.8724	0.3057	0.9061	0.9975	0.9934	0.9496
MTAD-GAT	ICDM'20	0.9719	0.6958	0.8465	0.8110	0.2819	0.8013	0.8822	0.4170	0.8211	0.9216	0.9922	0.8684
CAE-M	TKDE'21	0.9698	0.6958	<u>0.8465</u>	0.8102	0.2783	0.7917	0.8727	0.4118	0.9081	0.9670	0.9782	0.9368
GDN	AAAI'21	0.9698	0.6958	0.8463	0.8102	0.2913	0.7932	0.8778	0.4261	0.7171	0.9975	0.9925	0.8343
TranAD	VLDB'22	0.9977	0.6879	0.8438	0.8143	0.3959	0.8295	0.8998	0.5360	0.9051	0.9973	<u>0.9933</u>	<u>0.9490</u>
MUTANT	KBS'23	0.9543	0.6872	0.8296	0.7990	0.4025	0.8198	0.8947	0.5399	0.8865	0.9827	0.9754	0.9321
DCdetector	KDD'23	0.9634	0.6926	0.8387	0.8059	0.3869	0.8168	0.8852	0.5251	0.8679	0.9210	0.9727	0.8937
DTAAD	KBS'24	0.9697	0.6957	0.8462	0.8101	0.9017	0.3910	0.6950	<u>0.5455</u>	0.8463	0.9974	0.9892	0.9147
MSTIA-Net	Ours	0.9699	0.6958	0.8465	0.8103	0.4825	0.8296	0.9043	0.6101	0.9989	0.9974	0.9986	0.9981

- **MUTANT** [24]: A graph-based model that learns the correlations and each importance between variables via the temporal GCN and LSTM-based attention reconstruction.
- **DCdetector** [45]: An attention-based model that designs dual attention contrastive representation learning scheme.
- **DTAAD** [30]: An attention-based method that adopts dual TCN and Transformer models to capture the multi-scale temporal dependences with a meta-learning strategy.

1) *Analysis of Anomaly Detection*: Table III shows the performance comparison of state-of-the-art methods on the six complete datasets. Notably, all baseline methods use the same data splits for a fair comparison. From these tables, we have the following observations: MSTIA-Net outperforms other baselines on most MTS datasets, which demonstrates the superiority of our method in effectively leveraging the spatial-temporal dependencies from multiview feature patterns. The prediction-based methods, *i.e.*, LSTM-NDT and GDN obtain relatively poor results, which may be attributed to the inherent nature of prediction models in dealing with long-term sequences. Obviously, the reconstruction-based methods, *i.e.*, DAGMM, OmniAnomaly, MSCRED, MAD-GAN and USAD adopt the deep AE models and show a certain degree of performance gap. MSCRED gets satisfactory results compared to OmniAnomaly, which indicates the sufficiency of capturing the spatial-temporal patterns from multi-scale features. MAD-GAN and USAD achieve more promising results, reflecting the

positive effect of generative adversarial learning for temporal distributions. The graph-based methods, *i.e.*, MTAD-GAT and MUTANT both perform relatively worse than our MSTIA-Net. One possible explanation is that they only focus on modeling spatial-temporal dependencies but ignore the potential multi-cue interactions. The attention-based methods, *i.e.*, CAE-M, TranAD, DCdetector, and DTAAD gain competitive results, which confirms the advantage of modeling complete spatial-temporal dependencies through the Transformer structure.

2) *Analysis of Anomaly Diagnosis*: Table IV provides the diagnosis performance comparison of other baseline methods on the SMD for the sake of brevity. As shown in the table, our MSTIA-Net model remains significantly superior to others, verifying that our model is highly capable of MTS anomaly diagnosis and the parallel-attention mechanism enables it more suitable for the inter-correlated anomalies. Specifically, this task aims to distinguish the abnormality from each dimension, and our proposed MSTIA-Net can effectively learn the spatial-temporal dependencies from local-global interactive patterns to accurately diagnose multi-dimensional anomalies.

3) *Analysis of Model Efficiency*: We also further compared the model performance on the scale-restricted training sets and model computational cost. Table V shows the performance comparison under the data-constrained conditions, and implies that our model achieves the excellent results thanks to the meta-learning strategy. Fig. 5 illustrates the comparison results

TABLE IV

DIAGNOSIS PERFORMANCE COMPARISON OF OUR MSTIA-NET WITH BASELINE METHODS ON THE SMD

Method	H@100%	H@150%	N@100%	N@150%
LSTM-NDT	0.1920	0.2951	0.2586	0.3175
DAGMM	0.2617	0.4333	0.3153	0.4151
OmniAnomaly	0.2839	0.4365	0.3338	0.4231
MSCRED	0.2322	0.3469	0.2297	0.2962
MAD-GAN	0.3856	0.5589	0.4277	0.5292
USAD	0.3095	0.4769	0.3534	0.4515
MTAD-GAT	0.4912	0.5885	0.4126	0.4822
CAE-M	0.2530	0.4171	0.2047	0.3010
GDN	0.2276	0.3382	0.2921	0.3570
TranAD	0.3710	0.5509	0.4106	0.5204
MUTANT	0.3868	0.5662	0.4086	0.5192
DCdetector	0.3943	0.5724	0.4067	0.5176
DTAAD	0.4087	0.6235	0.4298	0.5565
MSTIA-Net	0.4796	0.6365	0.5145	0.5856

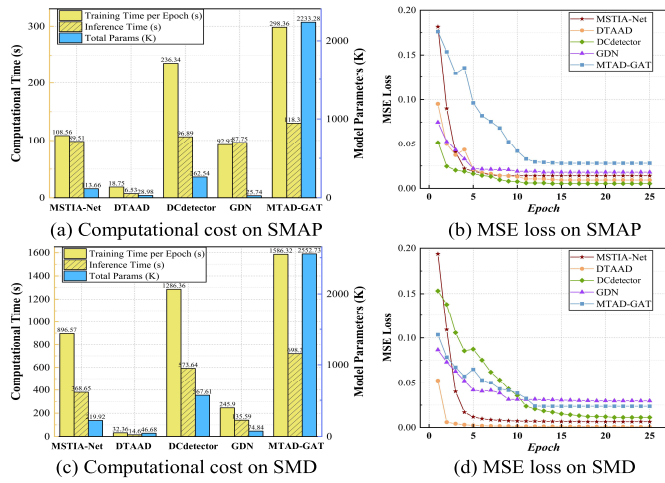


Fig. 5. Performance comparison of computational complexity and model convergence on the SMAP and SMD.

of computational complexity and model convergence, and demonstrates the efficiency of our method against suboptimal baseline methods. From the figures, we can find that our MSTIA-Net gets an acceptable range of training and inference time consumption and moderate number of model parameters. Due to the multi-node interactions of graph networks, the graph-based methods typically have higher time complexity than the attention-based methods. Notably, our method also exhibits a faster convergence rate, achieving better performance and model stability in fewer training epochs.

Overall, compared to the graph-based and attention-based methods, our significant strengths are that the MSTIA-Net can jointly learn spatial-temporal content-aware and relation-aware interactions in a multiview parallel-attention manner, thereby improving detection performance to varying degrees.

E. Ablation Analysis

To verify the contribution of each component in the MSTIA-Net model, we compared the performance of multiple ablation schemes by removing the relevant components as follows:

- **w/o Local / Global TCN**: We removed the effect of local or global temporal features by using single view feature.

TABLE V

PERFORMANCE COMPARISON OF OUR MSTIA-NET WITH BASELINE METHODS ON 20% OF THE TRAINING SETS

Method	MBA		MSL		SMAP	
	AUC*	F1*	AUC*	F1*	AUC*	F1*
LSTM-NDT	0.9616	0.9281	0.9519	0.7607	0.7006	0.5417
DAGMM	0.9670	0.9395	0.9605	0.8009	0.9880	0.8368
OmniAnomaly	0.9406	0.9216	0.9702	0.8423	0.9878	0.8130
MSCRED	0.9498	0.9107	0.9796	0.8231	0.9810	0.8049
MAD-GAN	0.9549	0.9191	0.9648	0.8189	0.9876	0.8467
USAD	0.9698	0.9426	0.9650	0.8191	0.9884	0.8380
MTAD-GAT	0.9689	0.9426	0.9783	0.8025	0.9815	0.8226
CAE-M	0.9617	0.9003	0.9837	0.7304	0.9893	0.8313
GDN	0.9672	0.9317	0.9415	0.8960	0.9888	0.8412
TranAD	0.9946	0.9897	0.9856	0.9171	0.9884	0.8936
MUTANT	0.9837	0.9718	0.9748	0.9239	0.9856	0.8860
DCdetector	0.9788	0.9636	0.9742	0.9086	0.9889	0.8992
DTAAD	0.9955	0.9912	0.9864	0.9212	0.9894	0.8996
MSTIA-Net	0.9958	0.9918	0.9867	0.9272	0.9896	0.8997

Method	SWaT		WADI		SMD	
	AUC*	F1*	AUC*	F1*	AUC*	F1*
LSTM-NDT	0.6689	0.4144	0.6636	0.0000	0.9562	0.6753
DAGMM	0.8420	0.8000	0.6496	0.0629	0.9844	0.8985
OmniAnomaly	0.8318	0.7432	0.7912	0.1016	0.9857	0.9351
MSCRED	0.8384	0.7921	0.6028	0.0412	0.9767	0.8003
MAD-GAN	0.8455	0.8011	0.5382	0.0936	0.8634	0.9317
USAD	0.8439	0.8088	0.7012	0.0734	0.9855	0.9214
MTAD-GAT	0.8460	0.8080	0.6268	0.0521	0.9799	0.6662
CAE-M	0.8459	0.7842	0.6110	0.0782	0.9570	0.9319
GDN	0.8391	0.8073	0.6122	0.0413	0.9812	0.7108
TranAD	0.8461	0.8093	0.6852	0.0698	0.9847	0.8794
MUTANT	0.8404	0.7890	0.7798	0.0892	0.9786	0.7081
DCdetector	0.8326	0.7968	0.7267	0.0796	0.9806	0.8727
DTAAD	0.8460	0.8087	0.7818	0.0977	0.9866	0.8941
MSTIA-Net	0.8472	0.8121	0.7455	0.0850	0.9876	0.9020

- **w/o STIA**: We eliminated the influence of Spatial-Temporal Interactions Aggregation (STIA) module without generating the content-aware representations.
- **w/o STCL**: We eliminated the influence of Spatial-Temporal Correlations Learning (STCL) module without generating the relation-aware representations.
- **w/o DACR**: We disabled the impact of Dual Attention-enhanced Contrastive Reconstruction (DACR) without cross-attention mechanism and contrastive learning.
- **w/o Spatial / Temporal MSA**: We disabled the influence of spatial or temporal self-attention Transformer by only using the single domain dependency embedding.
- **w/o LRBP**: We eliminated the impact of LRBP fusion strategy via a direct concatenation manner.
- **w/o Spatial / Temporal GAT**: We disabled the influence of spatial or temporal GAT layer by using the single domain contextual embedding.
- **w/o Callback**: We removed the effect of local reconstruction outputs without a skip-layer feedback mechanism.

Table VI shows the performance comparison of different ablation schemes, we can see that all variants consistently show inferior performance compared to overall MSTIA-Net model, which illustrates significant effectiveness of each component. In detail, the w/o Local or Global TCN performs worse than other schemes, demonstrating that modeling the multi-scale patterns of MTS data from multiview features can

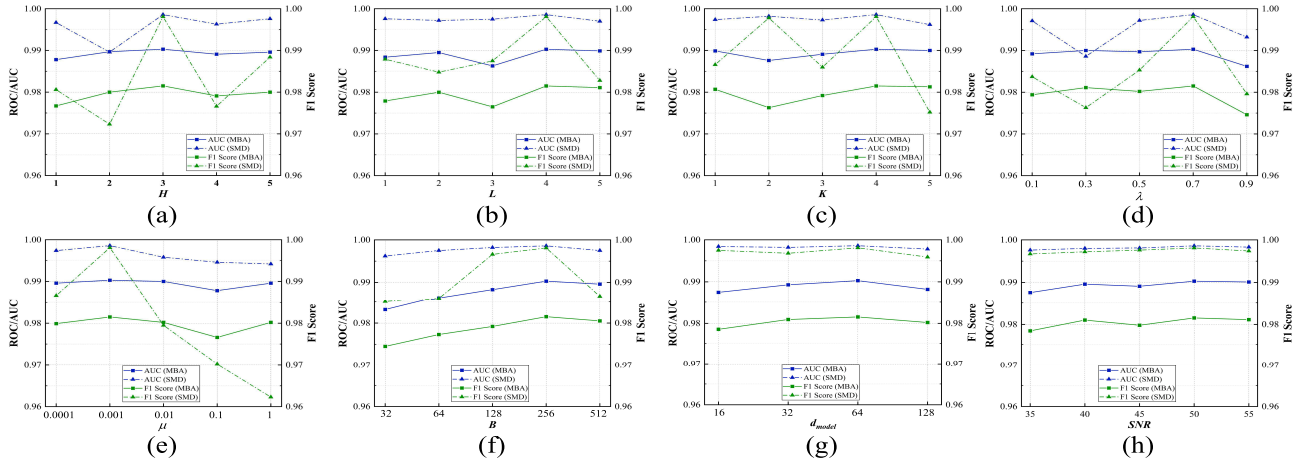


Fig. 6. Performance comparison with different values of parameters in our proposed MSTIA-Net on the MBA and SMD.

TABLE VI
PERFORMANCE COMPARISON OF DIFFERENT COMPONENT CONFIGURATIONS ON THE SIX DATASETS

Method	MBA		MSL		SMAP	
	AUC	F1	AUC	F1	AUC	F1
MSTIA-Net	0.9903	0.9815	0.9924	0.9466	0.9897	0.9051
w/o Local TCN	0.9896	0.9800	0.9449	0.7580	0.9895	0.8991
w/o Global TCN	0.9895	0.9798	0.9232	0.6739	0.9892	0.8985
w/o STIA	0.9894	0.9796	0.9677	0.8311	0.9892	0.8988
w/o STCL	0.9893	0.9796	0.9598	0.7979	0.9886	0.8944
w/o DACR	0.9896	0.9799	0.9913	0.9464	0.9893	0.8986
w/o Spatial MSA	0.9894	0.9798	0.9483	0.7543	0.9892	0.8991
w/o Temporal MSA	0.9891	0.9793	0.9314	0.6982	0.9891	0.8990
w/o LRBP	0.9893	0.9795	0.9419	0.7320	0.9891	0.8990
w/o Spatial GAT	0.9897	0.9802	0.9764	0.8708	0.9894	0.8958
w/o Temporal GAT	0.9894	0.9798	0.9336	0.6959	0.9892	0.8946
w/o Callback	0.9895	0.9800	0.9562	0.7838	0.9888	0.8963
Method	SWaT		WADI		SMD	
	AUC	F1	AUC	F1	AUC	F1
MSTIA-Net	0.8465	0.8103	0.9043	0.6101	0.9986	0.9981
w/o Local TCN	0.8463	0.8102	0.8978	0.5088	0.9985	0.9980
w/o Global TCN	0.8462	0.8101	0.8783	0.4992	0.9984	0.9979
w/o STIA	0.8460	0.8073	0.9004	0.5856	0.9963	0.9761
w/o STCL	0.8462	0.8101	0.5857	0.2616	0.9985	0.9974
w/o DACR	0.8462	0.8101	0.9015	0.5882	0.9932	0.9489
w/o Spatial MSA	0.8459	0.8075	0.8994	0.5835	0.9978	0.9949
w/o Temporal MSA	0.8462	0.8101	0.9010	0.5869	0.9964	0.9774
w/o LRBP	0.8462	0.8101	0.9024	0.5851	0.9985	0.9980
w/o Spatial GAT	0.8461	0.8086	0.8762	0.5209	0.9947	0.9618
w/o Temporal GAT	0.8463	0.8100	0.8937	0.5472	0.9978	0.9901
w/o Callback	0.8458	0.8026	0.8862	0.3912	0.9970	0.9973

effectively characterize temporal information. w/o STIA and w/o STCL both obtain unsatisfactory results, which implies the necessity of learning the content-oriented interactions based on parallel-attention mechanism and capturing the dynamic correlations via stacked GAT blocks. Notably, w/o DACR also gets poor performance due to the insufficient reconstruction means and missing auxiliary strategies for generating dual-stream outputs. Similar to the w/o Spatial or Temporal MSA, the w/o Spatial or Temporal GAT obviously drops compared to MSTIA-Net. The main reason is that both temporal and spatial information have certain content and correlation dependencies, which are beneficial to jointly construct fusion embedding. Limited by the capability of early fusion and implicit clues mining, both w/o LRBP and w/o Callback show poor

performance. The LRBP enhances interaction in the feature space to achieve refined encoding. MSTIA-Net achieves the best results, which can be attributed to the relevance of spatial-temporal patterns.

F. Parameter Sensitivity Analysis

To investigate the influence of spatial-temporal and local-global Transformer, we considered changing the number of attention heads (*i.e.*, H) by ranging H from 1 to 5. Fig. 6 (a) shows the performance with different numbers of heads in the Transformer, we can find that the results show significant fluctuations and the best performance is gained when $H = 3$. Possible explanation is that MHA mechanism not only assists spatial-temporal contents interaction and multi-view patterns integration, but also causes a certain degree of negative noise. Furthermore, we explored the impact of spatial-temporal GAT blocks by varying the number of GAT layers (*i.e.*, L) and MHA heads (*i.e.*, K), which both range from 1 to 5. Fig. 6 (b) and (c) show the results with different numbers of layers and heads in the STCL module. From the figures, our model achieves the optimal performance when $L = 4$ and $K = 4$, while also exhibits strong volatility. This can be attributed to the fact that the dynamic correlations of spatial-temporal domains can be effectively captured as the number of GAT layers increases, but excessively deep GAT layers lead to certain negative effects.

We also studied the influence of λ and μ in the total objective function $\mathcal{L}_{total}$, which are used to control reconstruction and cross-view contrastive loss terms. Fig. 6 (d) and (e) show the results with different values of trade-off parameters λ and μ , which are selected for the optimal parameters via a heuristic grid-search strategy with the same interval ranging from $\{0.1, 0.3, 0.5, 0.7, 0.9\}$ and $\{1e^r | r \in -4, \dots, 0\}$, respectively. From the figures, we can find that the best results are obtained when $\lambda = 0.7$ and $\mu = 1e^{-3}$. Nevertheless, μ exhibits stronger sensitive properties than λ , indicating the significance of cross-view contrastive learning strategy for the fusion embeddings.

Fig. 6 (f) and (g) show the results with different values of batch size (*i.e.*, B) and feature dimension (*i.e.*, d_{model}) by ranging from $\{32, 64, 128, 256, 512\}$ and $\{16, 32, 64, 128\}$. The optimal results are obtained when $B = 256$ and $d_{model} = 64$, and our model exhibits a certain degree of stability. In addition,

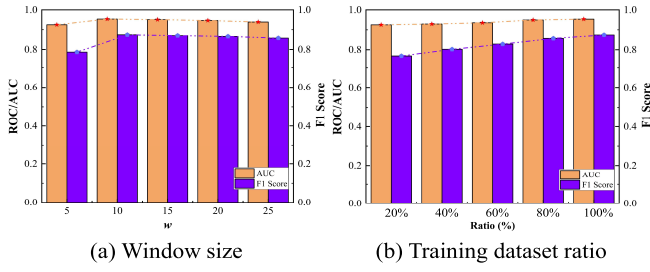


Fig. 7. Parameter sensitivity studies of the sliding window size and training dataset ratio on the six benchmark datasets.

we verify the impact of noise signals at different proportions on model performance in the simulated industrial scenarios. Fig. 6 (h) shows the performance with different SNR values of Gaussian white noise by ranging from $\{35, 40, 45, 50, 55\}$ dB, and our model has a certain degree of robustness against interference from noisy data of varying intensity.

We further explored the impact of sliding window size and training set ratio. Fig. 7 shows the averaged results with different values of w and Ratio (%), which are selected by varying from $\{5, 10, 15, 20, 25\}$ and $\{20\%, 40\%, 60\%, 80\%, 100\%\}$. The observations can be made: our model obtains the optimal results when $w = 10$, and the size of window has a certain impact on model performance. It may be that smaller windows are difficult to learn the global temporal feature, while local anomalies are hidden in the multiple points for larger windows. Obviously, under different conditions of limited-scale training set, our MSTIA-Net demonstrates strong model robustness.

G. Case Study

To intuitively understand the effect of MSTIA-Net model on anomaly detection, we conduct the following case analysis of our model and other schemes on the testing set of SMD.

1) *Anomaly Diagnosis Study*: Fig. 8 illustrates the visualization of anomaly diagnosis on multiple dimensions of testing samples with the MSTIA-Net and ablation methods, where the latter two only use single-view temporal feature. As shown in the Fig. 8, we can find that MSTIA-Net performs well overall on the MTS-AD task, which is able to diagnose the anomalous sequences based on the predicted anomaly scores. There are the associated abnormal points between dimensions #0 and #2 with relevant inter-relations, where anomalous behavior in one dimension may lead to a chain of events causing abnormalities in others. Similarly, the ablation schemes based on single-view features also exhibit specific detection effects in #3 and #4. It can be seen that the prediction results of local-global features are broadly consistent with anomaly labels, and discrepancy can be avoided by the point adjustment strategy. This validates that our model fully integrates the interactions of local-global temporal patterns, thus assisting in locating abnormal data.

2) *Multiview Interaction Study*: We also explored the impacts of multiview temporal features and dual cross-attention mechanism. Fig. 9 shows the T-SNE visualization of various temporal features. The MSTIA-Net with multiview feature obviously reduces the overlap between normal and abnormal sample embeddings. Both global and local temporal patterns also exhibit specific discriminability with clearer

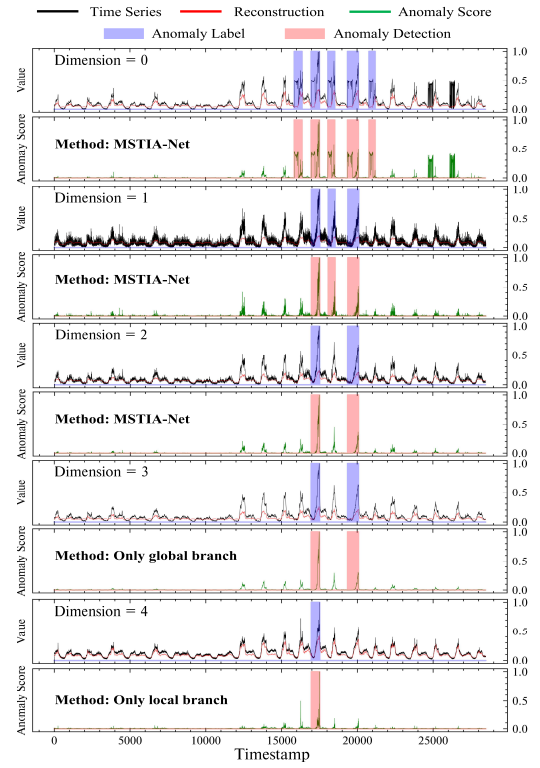


Fig. 8. Diagnosis visualization of anomaly prediction among multiple dimensions on the SMD testing set using MSTIA-Net and other ablation strategies. The time series with ground-truth values and reconstruction results are provided, and anomaly labels marked in purple are also presented for comparison. The predicted anomalous segments are highlighted in red.

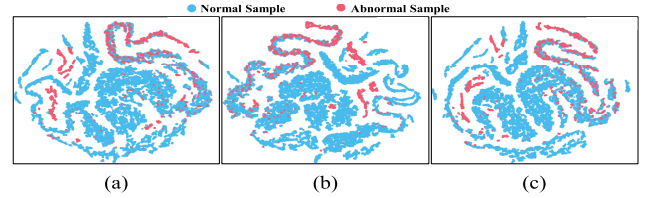


Fig. 9. T-SNE visualization of different temporal features before reconstruction learned by (a) multiview fusion, (b) only global-patterns, and (c) only local-patterns on the testing set of SMD. The dots with same color indicate the same label.

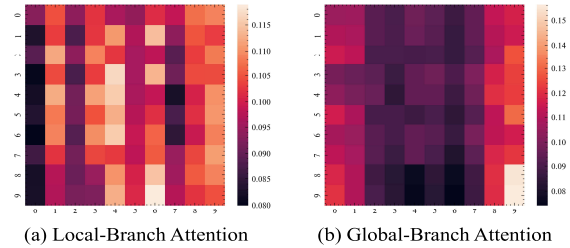


Fig. 10. Cross-Attention visualization of (a) Local-Branch and (b) Global-Branch attention matrices on the SMD. The coordinate axes represent the size of sliding window.

boundaries, we can find that multiview fusion features enhance discriminative representations in complex distributions. Fig. 10 shows the heatmap visualization of cross-attention matrix in the local and global perspectives, and implies that it has the ability to capture correlative interactions from different view features.

VI. CONCLUSION AND FUTURE WORK

In this study, we focus on jointly model the spatial-temporal dependencies of time-series in the distributed ICS, and propose a novel MTS-AD approach called MSTIA-Net, which consists of a spatial-temporal interactive encoder and a dual-stream contrastive decoder. Specifically, the STIA module is presented to learn spatial-temporal dependencies by a parallel-attention mechanism and aggregate content-aware representations with a low-rank fusion manner. In addition, the STCL module is designed to adaptively capture dynamic correlations by stacked GAT blocks. Conclusively, the cross-aware fusion embedding is further enhanced for generating dual-stream outputs through the DACR module. Extensive experimental results verify the superiority of our model and the effectiveness of each module.

In future work, we plan to exploit the powerful knowledge reasoning capability of Large Language Models (LLMs) to further explore more efficient solutions based on MSTIA-Net. Towards the demands of real industrial scenarios, our MSTIA-Net still has limitations in terms of high model complexity, so we will improve and deploy our model in a lightweight manner on the nuclear power industrial network for data management.

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